Software Engineering Lab 4: Vibe Coding

Student: Dilipan Y | **SRN:** PES1UG24AM092 | **Assigned Repo:** SETAPESU26/28_defender

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User Prompt (Task Assignment)

I have a Lab assignment in Software Engineering.
The handout pdf is in the current directory here are the instructions sent with it:
Upload the following to your Lab repo under Lab-4 folder:
a. Videos (before and after)
b. Updated Code
c. Chat history exported as a doc/pdf

Assigned Repo: https://github.com/SETAPESU26/28_defender

I have taken the before video. Now I want you to make the required changes and do the pushes to github after each change as per the requirements. Then tell me what to do and ask if you need anything.

AI Assistant Analysis & Clarification

Analyzed the Lab 4 handout and the Defender game codebase (SETAPESU26/28_defender).
Identified repository requirements and four mandatory code tasks:
1. **Task 1:** Fix radar blip positioning from screen-space to absolute world coordinates.
2. **Task 2:** Implement dynamic sky_color(wave) color shifts across waves.
3. **Task 3:** Implement on_humanoid_rescued(humanoid) with floating celebration score popup and temporary invulnerability shield boost.
4. **Task 4:** Implement bonus_life_threshold() awarding extra lives at 10,000 score milestones.

Confirmed target repo: https://github.com/Dilipan-Y-8488/SE-Lab directly under Lab-4/.

Task 1 Execution: Radar World-Space Mapping

Problem: The radar plotted blips using screen_x(x) relative to the player, causing all entities to cluster around the player's blip.
Solution: Replaced rx = self.screen_x(x) % VIEW_W with rx = int((x % WORLD_W) / WORLD_W * VIEW_W) to map true world coordinates [0, WORLD_W] across radar width [0, VIEW_W]. Added a scanner viewport indicator rectangle to display current camera view.
Git Commit: 62661d7 - *Lab 4: Task 1 - Fix radar blip positioning to world-space coordinates and add scanner viewport indicator*
Pushed to origin main.

Task 2 Execution: Dynamic Sky Atmosphere

Problem: sky_color(wave) was a no-op pass, resulting in default static black background.
Solution: Implemented sky_color(wave) returning cycling atmospheric color palettes (deep midnight navy, alien violet, twilight magenta, crimson nebula, toxic emerald) that evolve dynamically as waves progress.
Git Commit: 9071c51 - *Lab 4: Task 2 - Implement sky_color(wave) with dynamic atmospheric color shifting per wave*
Pushed to origin main.

Task 3 Execution: Humanoid Rescue Celebration & Shield

Problem: on_humanoid_rescued(humanoid) was an empty stub when player caught falling humanoids.
Solution: Implemented on_humanoid_rescued(humanoid) spawning floating animated +500 RESCUED! celebration popups and granting a 2.5-second energy shield / invulnerability ring protecting the player upon rescue.
Git Commit: b532705 - *Lab 4: Task 3 - Implement on_humanoid_rescued with floating celebration popup and shield boost*
Pushed to origin main.

Task 4 Execution: Bonus Life Threshold

Problem: bonus_life_threshold() returned None, disabling the bonus life feature.
Solution: Implemented bonus_life_threshold() returning 10000. Automatically grants an extra life every 10,000 points scored.
Git Commit: cc88409 - *Lab 4: Task 4 - Implement bonus_life_threshold to grant extra life every 10,000 points*
Pushed to origin main.

Verification & Delivery

Verified all mechanics with unit tests and headless Pygame execution. All commits successfully pushed to https://github.com/Dilipan-Y-8488/SE-Lab. Created convenient run_game.bat launcher for student gameplay recording.